

Lecture 2.1: RAS and MYC - The Oncogenic Powerhouses

Good morning, everyone. Today we're diving deep into the molecular world of oncogenes, focusing on two of the most important and well-studied families in cancer biology: RAS and MYC. These aren't just any oncogenes—they're the powerhouses that drive some of the most common and deadly cancers affecting humans today.

If you remember from our historical overview, the discovery that normal cellular genes could become cancer-causing oncogenes revolutionized our understanding of cancer. Today, we're going to explore how two particular families of these genes—RAS and MYC—have become central players in cancer development and why they've captured the attention of researchers and drug developers worldwide.

The RAS Family: Molecular Switches Gone Wrong

Let's start with the RAS family, consisting of three closely related genes: KRAS, NRAS, and HRAS. The story of RAS begins with studies of rat sarcoma viruses in the 1960s and 1970s, which led to the identification of these genes as the cellular versions of viral oncogenes. But what makes RAS so special, and why is it mutated in about one-third of all human cancers?

RAS as Molecular Switches: The GTPase Cycle

To understand how RAS functions and how it goes wrong in cancer, we need to appreciate its role as a molecular switch. RAS proteins belong to a large family called small GTPases, and like all GTPases, they function as binary switches that can be either "on" or "off."

Think of RAS as a sophisticated toggle switch in your cell's signaling circuitry. When RAS is bound to GTP—guanosine triphosphate—it's in the "on" position and actively transmits growth signals. When it's bound to GDP—guanosine diphosphate—it's in the "off" position and silent.

This switching mechanism is crucial for normal cell function. Cells need to respond to growth signals when appropriate, but they also need to turn off these signals when they're no longer needed. RAS proteins accomplish this through their intrinsic ability to hydrolyze GTP to GDP, effectively turning themselves off.

But here's where it gets interesting from a cancer perspective: this switch mechanism is tightly regulated by other proteins. Guanine nucleotide exchange factors, or GEFs, help load RAS with GTP, turning it on. GTPase-activating proteins, or GAPs, accelerate the hydrolysis of GTP to GDP, turning RAS off. In normal cells, this creates a carefully balanced system where RAS activity is precisely controlled.

Structural Biology: Understanding the Molecular Machinery

The three-dimensional structure of RAS proteins reveals how this switching mechanism works at the atomic level. RAS proteins have two critical structural regions: the GTP/GDP binding site and two flexible regions called Switch I and Switch II.

When GTP is bound, these switch regions adopt one conformation that allows RAS to interact productively with its downstream effector proteins. When GDP is bound, the switch regions adopt a different conformation that prevents these interactions.

The beauty of this system is its simplicity and reliability. The conformational changes are dramatic enough to completely alter which proteins can bind to RAS, yet they're triggered by the simple chemical difference between GTP and GDP.

The Cancer Connection: When Switches Get Stuck

Now, here's where cancer enters the picture. The vast majority of oncogenic RAS mutations occur at just three amino acid positions: codons 12, 13, and 61. These mutations don't prevent RAS from binding GTP or GDP, but they do something far more sinister—they impair RAS's ability to hydrolyze GTP to GDP.

The result is a RAS protein that gets stuck in the "on" position. Once it binds GTP, it can't efficiently convert it back to GDP, so it remains constitutively active, continuously sending growth signals to the cell.

This is a perfect example of how a small molecular change can have enormous biological consequences. A single amino acid substitution—say, glycine to valine at codon 12—is enough to convert a precisely regulated molecular switch into a constitutively active growth signal.

KRAS: The Clinical Champion

While all three RAS genes can cause cancer when mutated, KRAS stands out for its clinical significance. KRAS mutations are found in about 95% of pancreatic adenocarcinomas, 50% of colorectal cancers, and 30% of lung adenocarcinomas. This makes KRAS one of the most frequently mutated oncogenes in human cancer.

Why is KRAS so dominant? The answer lies partly in tissue-specific differences in RAS gene expression and function. Different tissues rely more heavily on different RAS family members for their normal growth and development. KRAS appears to be particularly important in the epithelial tissues that give rise to the most common solid tumors.

The clinical importance of KRAS mutations extends beyond their frequency. KRAS-mutant cancers tend to be particularly aggressive and difficult to treat. For decades, KRAS was considered "undruggable"—we knew it was driving these cancers, but we couldn't figure out how to inhibit it effectively.

RAS Signaling Networks: Multiple Pathways, Multiple Functions

When RAS is activated, it doesn't just turn on a single downstream pathway—it activates multiple signaling networks that collectively drive cancer development. Understanding these pathways is crucial for appreciating how RAS contributes to cancer and for developing effective therapies.

The RAF-MEK-ERK Pathway is perhaps the best characterized RAS effector pathway. When RAS-GTP binds to RAF kinases, it triggers a kinase cascade: RAF phosphorylates and activates MEK, which in turn phosphorylates and activates ERK. ERK then enters the nucleus and phosphorylates transcription factors that drive cell proliferation.

This pathway is particularly important for the "sustaining proliferative signaling" hallmark of cancer. Constitutively active RAS leads to continuous ERK signaling, which drives cells to proliferate regardless of external growth signals.

The PI3K-AKT-mTOR Pathway represents another major branch of RAS signaling. RAS-GTP can directly bind to and activate PI3K, which generates the lipid second messenger PIP3. This leads to activation of AKT, which has multiple pro-survival and growth-promoting effects.

AKT promotes cell survival by phosphorylating and inactivating pro-apoptotic proteins like BAD and FOXO transcription factors. It also activates mTOR, a master regulator of cell growth and metabolism. This pathway is crucial for the "resisting cell death" and metabolic reprogramming hallmarks of cancer.

The RAL-GEF Pathway is less well understood but equally important. RAS can activate RAL guanine nucleotide exchange factors, which activate RAL GTPases. These proteins are involved in cytoskeletal regulation, vesicle trafficking, and transcriptional control. RAL signaling contributes to multiple cancer hallmarks, including invasion and metastasis.

Therapeutic Breakthrough: Targeting KRAS G12C

For over three decades, RAS was considered undruggable. The protein lacks obvious binding pockets for small molecules, and its picomolar affinity for GTP makes competitive inhibition nearly impossible. But in recent years, we've witnessed a remarkable breakthrough in KRAS targeting.

The key insight came from recognizing that different KRAS mutations might have different vulnerabilities. The KRAS G12C mutation, found in about 13% of lung adenocarcinomas, creates a unique cysteine residue that can be targeted by covalent inhibitors.

Drugs like sotorasib and adagrasib work by covalently binding to this cysteine, locking KRAS G12C in its inactive, GDP-bound state. These drugs have shown remarkable clinical activity in KRAS G12C-mutant lung cancers, representing the first successful direct targeting of mutant KRAS.

This breakthrough has reinvigorated efforts to target other KRAS mutations and has validated the concept that even "undruggable" proteins can be targeted with sufficient ingenuity and persistence.

MYC: The Master Regulator

Now let's turn our attention to MYC, an oncogene family that's equally important but functions through completely different mechanisms. While RAS is a signaling protein that transmits growth signals, MYC is a transcription factor that directly controls gene expression.

The MYC family consists of three members: c-MYC, N-MYC, and L-MYC. Of these, c-MYC is the most broadly important and the most frequently altered in human cancers. MYC was originally discovered as the cellular homolog of the v-myc oncogene carried by avian myelocytomatosis virus, and it has since become one of the most studied oncogenes in cancer biology.

MYC as a Transcriptional Master Regulator

What makes MYC so powerful is its role as a master transcriptional regulator. MYC doesn't just control a few genes—it regulates approximately 15% of all human genes, making it one of the most influential transcription factors in the human genome.

MYC functions as part of a transcriptional complex that binds to specific DNA sequences called E-boxes. When MYC levels are high, it drives the expression of thousands of genes involved in cell cycle progression, metabolism, protein synthesis, and many other cellular processes.

This broad transcriptional program explains why MYC is so potent as an oncogene. Rather than affecting just one aspect of cell behavior, MYC simultaneously reprograms multiple cellular systems to promote rapid growth and proliferation.

Cellular Functions: A Growth-Promoting Program

MYC's target genes collectively orchestrate a cellular program focused on rapid growth and proliferation. Let me walk you through the major components of this program:

Cell Cycle Progression: MYC activates genes encoding cyclins, cyclin-dependent kinases, and other proteins that drive cells through the cell cycle. It also represses genes encoding cyclin-dependent kinase inhibitors that normally restrict cell cycle progression. The net result is that MYC-overexpressing cells cycle faster and are less responsive to growth-inhibitory signals.

Metabolic Reprogramming: MYC dramatically alters cellular metabolism to support rapid proliferation. It activates genes involved in glycolysis, glutaminolysis, and nucleotide biosynthesis—all processes that provide the energy and building blocks needed for rapid cell division. This metabolic reprogramming is a key component of the Warburg effect observed in cancer cells.

Protein Synthesis: MYC drives the expression of genes involved in ribosome biogenesis and protein translation. This makes sense because rapidly proliferating cells need to produce massive amounts of protein to support their growth and division.

Apoptosis Sensitization: Paradoxically, while MYC promotes growth and proliferation, it also sensitizes cells to apoptosis. This might seem counterintuitive, but it represents an important tumor suppressor mechanism. In normal cells, MYC activation in the absence of appropriate survival signals triggers apoptosis, eliminating potentially dangerous cells.

Oncogenic Mechanisms: How MYC Goes Wrong

Unlike RAS, which is typically activated by point mutations, MYC becomes oncogenic through several different mechanisms:

Gene Amplification is one of the most common ways MYC is activated in cancer. Many tumors contain extra copies of MYC genes, leading to overexpression of MYC protein. This is particularly common in certain types of breast cancer, where MYC amplification is associated with aggressive behavior and poor prognosis.

Chromosomal Translocations can place MYC under the control of highly active promoters, leading to inappropriate expression. The classic example is Burkitt lymphoma, where a translocation places c-MYC under the control of immunoglobulin gene regulatory sequences, leading to massive MYC overexpression in B cells.

Post-translational Stabilization represents another mechanism of MYC activation. Under normal conditions, MYC protein is rapidly degraded, giving it a half-life of only about 30 minutes. Various oncogenic signals can stabilize MYC protein, allowing it to accumulate to abnormally high levels.

The RAS-MYC Connection: Oncogenic Cooperation

One of the most important concepts in cancer biology is oncogene cooperation—the idea that multiple oncogenes often work together to drive cancer development. The relationship between RAS and MYC provides a perfect example of this cooperation.

RAS signaling directly affects MYC through multiple mechanisms. The ERK kinases activated by RAS can phosphorylate MYC, stabilizing it and enhancing its transcriptional activity. The PI3K-AKT pathway activated by RAS inhibits GSK-3 beta, a kinase that normally phosphorylates MYC and targets it for degradation.

This creates a powerful positive feedback loop: RAS activation leads to MYC stabilization, which drives the expression of genes that further enhance cell proliferation and survival. This cooperation helps explain why RAS and MYC are so frequently altered together in human cancers.

Metabolic Reprogramming: The MYC Metabolic Network

One of MYC's most important functions in cancer is orchestrating metabolic reprogramming. This isn't just about providing energy—it's about rewiring cellular metabolism to support the unique demands of rapid, uncontrolled proliferation.

Enhanced Glycolysis: MYC activates genes encoding glucose transporters and glycolytic enzymes, driving the Warburg effect. Even in the presence of oxygen, MYC-overexpressing cells preferentially use glycolysis for energy production. While this seems inefficient, it provides advantages for rapidly proliferating cells by generating biosynthetic precursors along with energy.

Glutaminolysis: MYC also drives glutamine metabolism, activating genes involved in glutamine uptake and conversion to alpha-ketoglutarate. Glutamine serves as both a nitrogen source for amino acid synthesis and a carbon source for the TCA cycle. In many cancers, glutamine becomes as important as glucose for supporting cell growth.

Nucleotide Biosynthesis: Perhaps most importantly for cancer, MYC activates the entire nucleotide biosynthesis pathway. Rapidly dividing cells need massive amounts of DNA and RNA precursors, and MYC ensures these are available by driving expression of enzymes in both purine and pyrimidine synthesis pathways.

This metabolic reprogramming represents a clear example of how oncogenes don't just drive proliferation—they anticipate and provide for the metabolic demands that rapid proliferation creates.

The Therapeutic Challenge: Targeting the "Undruggable"

MYC has long been considered even more challenging to target than RAS. As a transcription factor, MYC lacks the enzymatic active sites that provide obvious drug targets. Its interactions with DNA and other proteins involve large surface areas that are difficult to disrupt with small molecules.

Several strategies are being pursued to target MYC indirectly:

Targeting MYC Expression: Some approaches aim to reduce MYC expression using antisense oligonucleotides or small interfering RNAs. While challenging to deliver effectively, these approaches have shown promise in preclinical models.

Disrupting MYC-MAX Interaction: MYC requires a partner protein called MAX to bind DNA effectively. Small molecules that disrupt this protein-protein interaction could potentially inhibit MYC function.

Targeting MYC-Driven Metabolic Dependencies: Since MYC creates specific metabolic dependencies, these can potentially be exploited therapeutically. For example, MYC-overexpressing cells are often dependent on glutamine, making them vulnerable to glutaminase inhibitors.

Exploiting Synthetic Lethality: MYC activation creates vulnerabilities that can be exploited. For example, MYC-overexpressing cells are often more sensitive to inhibitors of protein synthesis or DNA repair.

Integration: RAS and MYC in Cancer Biology

As we conclude today's lecture, let's consider how RAS and MYC fit into the broader hallmarks of cancer framework we discussed earlier.

Sustaining Proliferative Signaling: Both RAS and MYC are central to this hallmark, but they contribute in different ways. RAS provides the signaling mechanisms that drive proliferation, while MYC provides the transcriptional program that executes it.

Evading Growth Suppressors: Both oncogenes help cells ignore growth-inhibitory signals. RAS mutations make cells independent of external growth factor signals, while MYC overexpression can overcome cell cycle checkpoints.

Reprogramming Energy Metabolism: This is particularly important for MYC, which orchestrates the metabolic changes needed to support rapid proliferation. RAS contributes through the PI3K-AKT-mTOR pathway, which regulates cellular metabolism.

Enabling Replicative Immortality: Both oncogenes contribute to overcoming senescence and enabling unlimited proliferative potential.

The cooperation between RAS and MYC illustrates an important principle: cancer typically results not from single oncogene activation, but from the coordinated action of multiple oncogenes that together drive the complex phenotypic changes we observe in cancer cells.

Clinical Implications and Future Directions

The study of RAS and MYC has already yielded important clinical insights and therapies. The recent success of KRAS G12C inhibitors proves that even challenging targets can eventually be drugged with sufficient effort and creativity.

Looking forward, several trends are likely to shape the future of RAS and MYC research:

Mutation-Specific Approaches: Just as KRAS G12C inhibitors target a specific mutation, we're likely to see more therapies designed for specific oncogene variants.

Combination Strategies: Given the cooperation between different oncogenes, combination therapies that target multiple pathways simultaneously may be more effective than single-agent approaches.

Biomarker Development: Understanding the downstream consequences of RAS and MYC activation will help us develop better biomarkers for selecting patients who will benefit from specific therapies.

Next time, we'll explore the other side of the oncogene story: tumor suppressor genes. We'll see how genes like TP53, APC, and RB normally function to prevent cancer, and how their loss contributes to cancer development. We'll also explore how the balance between oncogenes and tumor suppressors determines whether cells become cancerous.

Thank you for your attention, and I look forward to our discussion.